Real Analysis II
Problem Set 3: Sequences and Series

Exam style questions are marked with (*).

Question 1. (*) Prove the *Cauchy criterion for uniform convergence* of a series of functions.

Question 2. (*) Use the *Weierstrass M-test* to prove that the series of the following families of functions converge uniformly:

1. for $x \in \mathbb{R}$ let
$$f_n(x) = (-1)^n \frac{x^{2n}}{(2n)!};$$
2. for $x \in \mathbb{R}$ let
$$f_n(x) = \frac{n^2 + x^2}{n^4 + x^2};$$
3. for $x \in (-a, a)$ with $a \in \mathbb{R}$ let
$$f_n(x) = (-1)^n \frac{x^2 + n}{n^2}.$$

Question 3. (*) Use the *Lagrange remainder theorem* to estimate the error of convergence $E_N(x)$ of the following functions at the given points:

1. $f(x) = e^x$ when $x \in [0, 1]$ and $N = 6$;
2. $f(x) = \sin(x)$ when $x \in [-1, 1]$ and $N = 8$;
3. $f(x) = \arctan(x)$ at $x = 3$ and $N = 4$;

Question 4. How many terms in the Maclaurin series of $f(x) = \frac{1}{1-x}$ are needed to approximate $f(0.5)$ with error less than 0.0001?